

nano-size reaction region to accommodate all the fingers of all the manipulators necessary to have complete control of the chemistry

In his famous 1959 talk Nobel physicist Richard Feynman noted, “There’s plenty of room at the bottom.”, but Smalley jokes that there’s not that much room. Manipulator fingers on the hypothetical self-replicating nanobot are not only too fat; they are also too sticky: the atoms of the manipulator hands will adhere to the atom that is being moved. So it will often be impossible to release this minuscule building block in precisely the right spot.

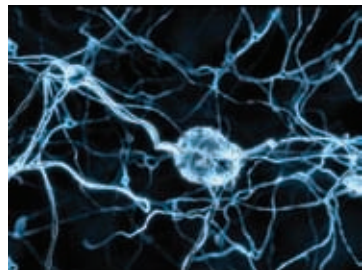
Smalley concludes that self-replicating, mechanical nanobots are simply not possible in our world. To put every atom in its place—the vision articulated by some nanotechnologists— would require magic fingers. Such a nanobot will never become more than a futurist’s daydream.

Rupturing the Nanotech Rapture

The second paper we will be discussing was written by Richard A.L. Jones, who is a professor of physics at the University of Sheffield, in England, and senior nanotechnology advisor for the UK government’s physical sciences and engineering funding agency. This paper entitled “Rupturing the Nanotech Rapture” was printed as a part of IEEE Spectrum’s “Special Report: The Singularity” in June 2008. In this paper, Jones has presented an interesting critique of the room-temperature molecular nanomachinery and that nanobots, if possible, would be more “bio” than “bot”.

Jones begins by attacking the expansive view of molecular nanotechnology, as touted by K. Eric Drexler in his 1992 book *Nanosystems*. In this book, Drexler envisaged nanostructures built from the strongest and stiffest materials available, using the rational design principles of mechanical engineering. The fundamental building blocks of this paradigm are tiny, rigid cogs and gears, analogous to the plastic pieces of a Lego set. The gears would distribute power from nanoscale electric motors and be small enough to assist in the task of attaching molecules to one another.

Even if we can assume that an object’s structure could easily be reduced to its molecular blueprint, Jones says that the first order of business would be figuring out how to translate macroscale manufacturing methods into nanoscale manipulations. For example, let’s say you wanted a new pancreas. Your first major challenge stems from the fact that a single



Special Report: The Singularity